

Department of Computer Science and Engineering
BRAC University

Set A

Examination: Midterm
Duration: 1 hour 30 minutes

Semester: Spring 2026
Full Marks: 30

CSE 423: Computer Graphics

Name:	ID:	Section:
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*Answer the following questions. Understanding the questions is part of the examination.
The numbers on the right margin indicate marks.*

Question 1 [CO1]

- a. In Minecraft, the world is a grid where players build using square blocks placed at integer coordinates. You want to build a straight stone path defined by the equation $\frac{x}{7} + \frac{y}{4} = 1$. The path starts exactly where this line crosses the x-axis and ends where it crosses the y-axis. **Determine** the exact coordinates where you will place the **intermediate blocks** if you were to use the Midpoint Line Algorithm. Show the necessary calculations in a table. 6
- b. One of your friends praises your choice of the **Midpoint Line Algorithm** over the **DDA** algorithm. They claim that the Midpoint algorithm (i) *is more accurate than DDA* and (ii) *will remove the staircasing (jagged edges) from the path*. Do you agree with these statements? Why or why not? **Explain** briefly. You must explicitly address both parts of their claim. 2+2

Question 2 [CO2]

Ross is working on a line for an architecture project. The equation of the line is given in parametric form as $P(t) = (-8t + 1, 6t + 5)$

Suddenly, Joey comes along and asks Ross, “Can you draw the segment of the line **P(t)** that starts at **t = -0.5** and ends at **t = 2** in your display window? I bet you can’t!”

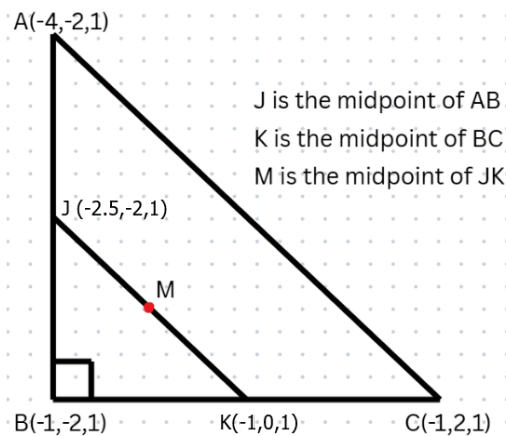
Ross begins working. He needs to draw the line in his rectangular display window, where the top-left corner is **C₁(-7, 15)** and the bottom-right corner is **C₂(18, -2)**. The edges of the window are parallel to the x-axis and y-axis. After much effort, Ted finds that Joey is correct!

- a. **Identify** the coordinates (x, y) of the starting and ending points of the line segment mentioned by Joey. 2

- b. Joey claimed that Ross cannot fully draw the line segment within the display window. **Justify** his claim mathematically using region codes (outcodes) only. **2**
- c. **Determine** the coordinates of the point(s) where the line segment intersects the window's edge(s) using a **region code-based** algorithm (with region codes in **TBRL** order). Also, **determine** the t-value(s) of the point(s) using the line's equation **P(t)**. **5+1**

Question 3 [CO3]

- a. Sevin is solving a complex puzzle left by an ancient master. The puzzle is written on a stone tablet along with the diagram below. **6**



Upon close inspection, the puzzle states that the Triangle ABC has gone through a few different transformations. Firstly, it was rotated 30 degrees counterclockwise. After that, the triangle is scaled with the following factors $S_x = 2$, $S_y = 12$, and $S_z = 1$. Finally, the triangle was sheared along the z-axis according to the transformation: $(x', y') = (x + z, y + 2z)$. All transformations are performed with respect to **point M** shown in the diagram.

Show the composite transformation matrices in the correct sequence and **calculate** the final coordinates of **point A** of the triangle after the puzzle is fully solved.

- b. **Derive** the 4×4 homogeneous transformation matrix for rotating a 3D point about the **Y-axis** with respect to an arbitrary center of rotation (p, q, r). **4**